Duality functor

We say that two functors $F_1, F_2 : \mathcal{C} \to \mathcal{D}$ are isomorphic if for any object $M \in \mathrm{Ob}\,\mathcal{C}$, there exists isomorphism $\eta_M : F_1(M) \to F_2(M)$ such that for any morphism $f : M \to N$ in $\mathcal{C}$ the following diagram commutes:

$$
\begin{tikzcd}
F_1(M) \arrow[r, "F_1(f)"] \arrow[d, "\eta_M"'] & F_1(N) \arrow[d, "\eta_N"] \\
F_2(M) \arrow[r, "F_2(f)"] & F_2(N)
\end{tikzcd}
$$

Denote $F_1 \simeq F_2$

We say that functor $F : \mathcal{C} \to \mathcal{D}$ is an equivalence of categories if there exists $G : \mathcal{D} \to \mathcal{C}$ s.t.

$F \circ G : \mathcal{D} \to \mathcal{D}$ and $G \circ F : \mathcal{C} \to \mathcal{C}$ are both

$F \circ G \simeq id_{\mathcal{D}}$ $\quad$ $G \circ F \simeq id_{\mathcal{C}}$

G is called quasi-inverse of F

A contravariant functor F which has quasi-inverse is called duality functor

Apply this to quivers . first Q^{op}

$Q^{op}_i = Q_i$, $\alpha: i \to j \in Q \Rightarrow \alpha^{op}: j \to i \in Q^{op}$

$\forall \alpha \in Q$, define α^{op} : $s(\alpha^{op}) = t(\alpha)$, $t(\alpha^{op}) = s(\alpha)$

we will use notations : $P_Q(i)$, $P_{Q^{op}}(i)$

D : $\mathrm{Hom}(-, k)$: $\mathrm{Rep}\,Q \longrightarrow \mathrm{Rep}\,Q^{op}$, contravariant functor

$M = (M_i, \varphi_\alpha)$ $\quad DM = (DM_i, D\varphi_\alpha)$ $\quad DM_i = \mathrm{Hom}(M_i, k)$

$D\varphi_\alpha = \mathrm{Hom}(\varphi_\alpha, k)$

$\alpha: i \to j$ in Q $\quad \varphi_\alpha: M_i \longrightarrow M_j$

$\alpha^{op}: j \to i$ in Q^{op} $\quad \mathrm{Hom}(M_j, k) \xrightarrow{D\varphi_\alpha} \mathrm{Hom}(M_i, k)$

$f: M_j \to k$ $\quad D\varphi_\alpha(f) = f \circ \varphi_\alpha$

$$\begin{tikzcd} M_i \arrow[r, "\varphi_\alpha"] \arrow[dr, dotted, "D\varphi_\alpha(f)"'] & M_j \arrow[d, "f"] \\ & k \end{tikzcd}$$

For morphism $h: M \to N$

$Dh: DN \longrightarrow DM$

$\mathrm{Hom}(N_i, k) \xrightarrow{\mathrm{Hom}(h_i, k) = h_i^*} \mathrm{Hom}(M_i, k)$

$\psi: N_i \to k$ $\quad h_i^*(\psi) = \psi \circ h_i$

In the same way we may define

D^{op} : $\mathrm{Hom}(-, k)$: $\mathrm{Rep}\,Q^{op} \longrightarrow \mathrm{Rep}\,Q$ (actually $D = D^{op}$)

hence $D^{op} \circ D$: $\mathrm{Rep}\,Q \longrightarrow \mathrm{Rep}\,Q$ (actually D^2)

$\mathcal{D}^2 M_i = \mathrm{Hom}(\mathrm{Hom}(M_i, k), k) \simeq M_i$

$\mathcal{D}^2 \simeq \mathrm{id}_{\mathrm{Rep}Q}$

$\mathcal{D}$ has quasi-inverse which is $\mathcal{D}^{op}$

Example: $\underset{1}{\bullet} \longrightarrow \underset{2}{\bullet} \longleftarrow \underset{3}{\bullet}$ Q

indecomposable projectives are $\begin{smallmatrix}1\\2\end{smallmatrix}$ $\quad 2 \quad$ $\begin{smallmatrix}3\\2\end{smallmatrix}$

Q^{op}: $\underset{1}{\bullet} \longleftarrow \underset{2}{\bullet} \longrightarrow \underset{3}{\bullet}$

$D(k \xrightarrow{1} k \leftarrow 0) = k \xleftarrow{1} k \to 0$ $\quad D(P_Q(1)) = I_{Q^{op}}(1)$

$D(0 \to k \leftarrow 0) = 0 \leftarrow k \to 0$ $\quad D(P_Q(2)) = I_{Q^{op}}(2)$

$D(0 \to k \xleftarrow{1} k) = 0 \leftarrow k \xrightarrow{1} k$ $\quad D(P_Q(3)) = I_{Q^{op}}(3)$

Proposition: Duality restricts to

$\mathcal{D}: \mathrm{Proj}\,Q \longrightarrow \mathrm{Inj}.\,Q^{op}$ $\quad D(P_Q(i)) = I_{Q^{op}}(i)$

Consider functor: $\mathrm{Hom}(-, X)$

contravariant and if you apply it to $\mathrm{Rep}Q$

$\mathrm{Hom}(-, X)$ has vector space structure.

If $X = A$ the free representation of Q, i.e. $A = \bigoplus_{i \in Q_0} P(i)$

Then $\mathrm{Hom}(-, A): \mathrm{Rep}Q \longrightarrow \mathrm{Rep}Q^{op}$

Let $M = (M_i, \varphi_\alpha) \in \mathrm{Rep}Q$. put $N = (N_i, \psi_{\alpha^{op}})$

We have $N_i = \mathrm{Hom}(M, P(i))$

and for $\psi_{\alpha^{op}} : \mathrm{Hom}(M, P(j)) \to \mathrm{Hom}(M, P(i))$

$\alpha : i \to j$

$\alpha^{op} : j \to i$

$$\begin{tikzcd} M \arrow[r, "f"] \arrow[dr, "\alpha \circ f"'] & P(j) \arrow[d, "\alpha"] \\ & P(i) \end{tikzcd}$$

$$\psi_{\alpha^{op}}(f) = \alpha \circ f \in \mathrm{Hom}(M, P(i))$$

We got that $N = (N_i, \psi_{\alpha^{op}})$ in $\mathrm{Rep}\, Q^{op}$

to show that $\mathrm{Hom}(-, A)$ is a functor we need to show that for any $f : M \to M'$

$\alpha : i \to j$

$\alpha^{op} : j \to i$

$$\begin{tikzcd} \mathrm{Hom}(M', P(j)) \arrow[r, "{\mathrm{Hom}(f, P(j)) = f_j^*}"] \arrow[d, "\psi'_{\alpha^{op}}"] & \mathrm{Hom}(M, P(j)) \arrow[d, "\psi_{\alpha^{op}}"] \\ \mathrm{Hom}(M', P(i)) \arrow[r, "{\mathrm{Hom}(f, P(i)) = f_i^*}"'] & \mathrm{Hom}(M, P(i)) \end{tikzcd}$$

$\mathrm{Hom}(f, P(i))$:

$$\begin{tikzcd} M \arrow[r, "f"] \arrow[dr, dashed, "g \circ f"'] & M' \arrow[d, "g"] \\ & P(i) \end{tikzcd}$$

Let $\theta \in \mathrm{Hom}(M', P(j))$

$\psi_{\alpha^{op}}$

$$\psi_{\alpha^{op}} f_j^*(\theta) = \psi_{\alpha^{op}}(\theta \circ f) = \alpha \circ \theta \circ f$$

$$f_i^* \psi'_{\alpha^{op}}(\theta) = f_i^*(\alpha \circ \theta) = \alpha \circ \theta \circ f$$

$\Rightarrow$ $\mathrm{Hom}(-, A)$ is functor from $\mathrm{Rep}\, Q$ to $\mathrm{Rep}\, Q^{op}$

$\nu = D\,\mathrm{Hom}(-,A) : \mathrm{Rep}Q \to \mathrm{Rep}Q$

$$\mathrm{Rep}Q \xrightarrow{\mathrm{Hom}(-,A)} \mathrm{Rep}Q^{op} \xrightarrow{D} \mathrm{Rep}Q.$$

covariant

Nakayama functor.

First consider $\mathrm{Hom}(-,A)$ on indecomposible projectes

let $M = \mathrm{Hom}(P_Q(i), A)$ $\quad M_j = \mathrm{Hom}(P_Q(i), P_Q(j))$

the space of all paths from j to i.

M_j has basis consisting of all paths from i to j in Q^{op}

Moreover any arrow $h \xrightarrow{\alpha} j$ in Q, we have.

$\varphi_{\alpha^{op}} : M_j \to M_h$

all paths from i to j in Q^{op} $(i|\cdots|j)$ $\quad$ all paths from i to h in Q^{op} $(i|\cdots \alpha^{op}|h)$

$M = \mathrm{Hom}(P_Q(i), A)$ is Indecomposable projective Q^{op}- representation.

$P_{Q^{op}}(i)$

remark projectives in i of Q^{op}

in each vertex vector space spanned by all paths in Q^{op}

starting from i to j. $\varphi_{\alpha^{op}} : P(i)_j \to P(i)_k$

$\Rightarrow \quad Hom(-,A)(P_Q(i)) = P_{Q^{op}}(i)$

$D(P_Q(i)) = I_{Q^{op}}(i)$

The restriction of $Hom(-,A)$: $proj\,Q \longrightarrow proj\,Q^{op}$

quasi-inverse functor $Hom_{Q^{op}}(-,A^{op})$

$A^{op} = \bigoplus_{i \in Q_0^{op}} P_{Q^{op}}(i)$

prop: restriction of ν on $proj\,Q$ in an equivalence of categories $proj\,Q \to inj\,Q$ with quasi-inverse given by $\nu^{-1} = Hom(DA^{op},-)$ $inj\,Q \to proj\,Q$.